

SYRACUSE HANCOCK, NY, USA

WMO: 725190

Lat: 43.111N

Lon: 76.104W

Elev: 126

StdP: 99.82

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14771

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.6	-15.5	-22.6	0.5	-17.8	-19.8	0.7	-14.6	13.4	-1.0	12.0	-4.3	2.9	90	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	31.7	22.8	30.2	21.9	28.8	21.1	24.1	29.5	23.2	28.1	22.4	27.0	4.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.4	26.9	21.5	16.4	25.9	20.7	15.6	25.2	72.9	29.7	69.4	28.1	66.2	27.0	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	9.2	8.3	-22.8	34.2	4.0	1.9	-25.7	35.5	-28.1	36.6	-30.3	37.7	-33.3	39.0		
			-23.1	25.8	4.0	1.1	-26.0	26.6	-28.3	27.2	-30.5	27.8	-33.4	28.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-4.4	-3.4	1.1	8.1	14.7	19.6	22.3	21.4	17.5	11.0	4.8	-0.7
	DBStd	10.56	6.84	5.89	5.91	5.06	4.59	3.73	3.02	2.94	3.97	4.45	4.93	5.28
	HDD10.0	1749	448	374	282	94	9	0	0	0	1	41	167	333
	HDD18.3	3613	705	607	534	310	132	29	3	7	63	229	404	590
	CDD10.0	1536	1	1	7	37	155	289	381	354	225	73	12	1
	CDD18.3	358	0	0	0	2	20	68	125	103	37	3	0	0
	CDH23.3	3046	0	0	7	42	215	605	1083	774	296	24	0	0
	CDH26.7	893	0	0	1	10	55	184	342	220	79	2	0	0
Wind	WSAvg	3.7	4.3	4.3	4.2	4.1	3.5	3.2	3.0	2.8	3.0	3.5	3.9	4.2
Precipitation	PrecAvg	957	63	54	75	89	87	82	89	86	89	77	88	77
	PrecMax	1342	161	114	137	196	199	261	203	151	187	169	147	133
	PrecMin	706	27	16	26	36	15	25	19	26	19	5	32	36
	PrecStd	150	28	23	27	38	43	44	46	32	37	38	28	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.3	15.9	23.1	28.3	30.7	32.9	33.6	33.4	31.8	26.2	20.7	17.0
		MCWB	13.1	11.2	14.9	17.2	20.7	23.7	24.1	23.4	22.7	18.9	14.9	12.3
	2%	DB	11.4	10.7	16.9	23.6	28.2	30.8	31.8	30.8	28.9	23.2	18.0	11.9
		MCWB	9.0	7.0	11.0	14.5	19.4	22.0	23.4	22.2	21.3	17.3	12.8	8.8
	5%	DB	7.6	7.5	13.4	20.1	25.8	28.8	30.1	29.1	26.7	21.0	15.8	9.1
		MCWB	5.5	4.5	8.7	12.6	18.0	20.9	22.2	21.4	20.4	16.3	11.7	6.8
	10%	DB	4.6	4.7	10.1	17.1	23.2	26.9	28.5	27.5	24.6	18.5	12.8	6.4
		MCWB	2.6	2.3	6.2	11.2	16.4	20.0	21.3	20.8	19.2	14.3	9.5	4.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	11.9	15.6	18.8	22.6	24.7	25.8	25.0	24.1	20.6	15.8	13.1
		MCDB	16.0	14.9	21.7	26.1	28.9	31.0	31.8	30.5	28.7	23.2	19.3	16.4
	2%	WB	8.9	7.4	12.0	15.8	20.6	23.3	24.3	23.6	22.6	18.5	14.0	9.5
		MCDB	11.1	10.2	16.0	21.5	26.3	28.9	29.8	28.2	27.4	22.3	16.9	11.3
	5%	WB	5.6	4.7	9.1	13.8	19.1	22.1	23.3	22.7	21.2	16.7	11.9	6.9
		MCDB	7.3	7.2	13.0	18.9	24.2	27.1	28.3	27.0	25.3	20.3	15.1	9.1
	10%	WB	2.8	2.5	6.5	11.5	17.6	20.9	22.4	21.7	20.0	14.9	9.8	4.5
		MCDB	4.4	4.5	10.1	16.2	22.1	25.2	27.0	26.0	23.6	18.1	12.6	6.2
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.4	9.1	10.8	11.4	10.7	10.5	10.4	10.7	9.3	8.1	7.0
		MCDBR	11.0	12.3	14.4	16.7	14.6	13.2	12.1	12.3	13.2	13.0	12.6	10.5
		MCWBR	9.3	9.6	9.4	9.4	7.6	6.4	5.6	5.6	6.6	7.9	8.8	8.5
	5% WB	MCDBR	10.6	11.4	13.6	14.8	12.9	11.2	10.4	10.3	11.3	11.4	11.4	10.0
		MCWBR	9.2	9.4	9.6	9.4	7.3	6.3	5.5	5.2	6.1	7.5	8.7	8.6
Clear-Sky Solar Irradiance	taub	0.296	0.310	0.335	0.364	0.393	0.437	0.445	0.429	0.377	0.356	0.315	0.291	
	taud	2.355	2.330	2.385	2.367	2.329	2.231	2.215	2.245	2.377	2.435	2.502	2.455	
	Ebn at Noon	835	885	907	902	881	839	827	827	850	824	815	815	
	Edh at Noon	80	98	106	116	124	138	139	130	106	88	70	67	
All-Sky Solar Radiation	RadAvg	1.32	2.15	3.35	4.40	5.34	5.66	5.80	5.16	4.10	2.48	1.56	1.02	
	RadStd	0.14	0.26	0.29	0.51	0.53	0.49	0.48	0.28	0.38	0.26	0.20	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+60	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+84	N/A

Nomenclature: See separate page